Quantized Adapters for Open Models

Okafor, Chen et al. · 2024

Fictional sample paper generated for the ClaimLens demo.

Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Our code and quantized checkpoints are publicly available. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models.

Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section.

We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps.

This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models.

Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration.

We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one.

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Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant.

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We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. We train LoRA adapters on top of a frozen 4-bit quantized base model. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

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Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. We adapt base models from 7B to 65B parameters. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

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Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. Across all eight benchmarks in Table 3, our 4-bit adapters match 16-bit full fine-tuning to within run-to-run variance. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Table 3 lists results on an eight-task benchmark suite. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

Table 3

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report accuracy for every task. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Our 4-bit adapters match 16-bit full fine-tuning on every benchmark in Table 3. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split.

We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Rank has little effect on final quality once all layers are adapted. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

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Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

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The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. We leave mathematical reasoning benchmarks to future work. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split.

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We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix.

Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split.

Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced.

All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds.